Bayesian Statistics I: Foundations

ECON 8310: Business Forecasting — Lecture 10

Department of Economics

University of Nebraska at Omaha

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Lecture Outline

Why Bayesian Thinking?

A different answer to the question of what a forecast should tell you.

Frequentist vs. Bayesian: What Is Random?

The whole difference is which thing you treat as uncertain.

	Frequentist	**Bayesian**
The parameter	A fixed, unknown constant	A quantity you are uncertain about
The data	Random — one draw of many possible samples	Fixed — it is what you observed
The answer	A point estimate and a standard error	A full distribution over values
Prior belief	No formal place for it	Stated explicitly, and updated

Everything else follows from that first row. If the parameter is fixed, you can only describe how your *estimate* would vary across hypothetical repeated samples. If the parameter is uncertain, you can describe your uncertainty about *it* directly — which is almost always the question someone actually asked.

What This Buys You in a Meeting

A 95% confidence interval of $[0.02,\ 0.08]$ does *not* mean there is a 95% chance the parameter lies in that range. It means that intervals built this way would contain the true value 95% of the time across repeated samples. That is a statement about the procedure.

A 95% **credible** interval of $[0.02,\ 0.08]$ does mean there is a 95% probability the parameter lies in that range, given your model and data.

> That is not a technicality. It is the difference between a sentence you can say out loud to a decision-maker and one you have to keep correcting.

And because the answer is a full distribution, you can ask questions of it directly: *What is the probability the promotion lifts demand by more than 5%? What is the chance we stock out if we order 400 units?* Those are the questions a planner has, and a point estimate with a standard error does not answer them.

The cost is that you must state a prior, and defend it. That is the rest of this lecture.

Bayes' Theorem

One equation, and it is a bookkeeping rule rather than a philosophy.

Bayes' Theorem (Gelman et al. 2013)

The update rule

$$\underbrace{p(\theta \mid \mathcal{D})}_{\text{posterior}} = \frac{\overbrace{p(\mathcal{D} \mid \theta)}^{\text{likelihood}} \cdot \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(\mathcal{D})}_{\text{evidence}}}$$

Read it as a sentence. **What I believe after seeing the data** equals **how well each value explains the data**, times **what I believed beforehand**, rescaled to sum to one.

The denominator is only that rescaling — it does not change the *shape*. So the working form is:

$$\text{posterior} \;\propto\; \text{likelihood} \times \text{prior}$$

You will not compute this by hand, and this course never asks you to. You are responsible for choosing the prior, checking the sampler worked, and reading the posterior.

The Update, in Pictures

Take a concrete case: estimating a conversion rate θ from 200 trials.

Prior	A curve over θ from 0 to 1, showing what you thought before. Broad if you know little; peaked if you have history.
Likelihood	A second curve, from the data alone: 30 successes out of 200 peaks near 0.15.
Posterior	Multiply the two curves point by point and rescale. It sits *between* them, closer to whichever was more confident.

Our numbers: the prior is centered on 0.10, the data say 0.15, and the posterior mean lands at **0.14** — between them, nearer the data, because 200 observations outweigh a prior worth about 50.

More data sharpens the likelihood, so the posterior moves toward it and the prior matters less. With little data the prior does most of the work — a feature when it is defensible, a problem when it is not.

Prior Distributions

The part you have to justify, and the part people worry about.

Priors You Will Actually Use

A prior is a distribution over a parameter, chosen to match what that parameter *can* be.

Prior	**Support**	**Use it for**
$\theta \sim \text{Beta}(\alpha, \beta)$	$[0, 1]$	Rates and proportions — conversion, click-through, defect rate
$\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$	all reals	Regression coefficients, trends, anything that can be negative
$\theta \sim \text{Exp}(\lambda)$	$[0, \infty)$	Standard deviations, waiting times, anything that must be positive

The support is the first thing to get right. Putting a Normal prior on a standard deviation allows negative values, which are impossible — the sampler will struggle and the model may not be wrong so much as incoherent.

Beta$(5, 45)$ *has mean* $5/50 = 0.1$ — *a prior centered on a 10% rate, worth about 50 observations of prior evidence.*

Choosing a Prior You Can Defend

The standard objection to Bayesian work is that the prior is arbitrary. The standard answer is that you should not be choosing arbitrary priors.

Weakly informative	Rules out the absurd, stays agnostic about the plausible. A Normal(0, 1) on a standardized coefficient says "probably not larger than about 2," which is nearly always true and nearly never controversial. **This is the default.**
Informative	Encodes real prior evidence — last year's conversion rate, a published elasticity. Defensible when you can name the source.
Flat / "uninformative"	Sounds neutral, often is not, and can be improper. Avoid it as a reflex choice.

The honest framing: a prior is an assumption stated in public, where it can be argued with. Frequentist analyses contain assumptions too — functional form, variable selection, sampling scheme — they are just not written as distributions.

Posterior Inference with PyMC

Sampling, the code, and how to tell whether the answer is trustworthy.

MCMC: The Big Idea

For any model more complicated than a textbook example, the rescaling denominator in Bayes' theorem cannot be computed. So we do not compute the posterior — we **draw samples from it**, and work with the samples.

The picture to hold. You are exploring a hilly landscape in the dark, where height is posterior probability. You take a step, and if the ground is higher you move; if it is lower you usually move back. Over thousands of steps you spend most of your time on the high ground.

The record of where you walked *is* the answer. A histogram of your visited positions has the shape of the posterior, without ever having computed it.

> **The samples are the answer.** Posterior mean, credible interval, and the probability of any event you care about are all just summaries of that list of visited values — counted, not derived.

PyMC's default sampler is NUTS, which takes smarter steps than the random walk above. The intuition is unchanged.

PyMC: A First Model

Estimating a conversion rate — 30 successes in 200 trials

```
import pymc as pm

with pm.Model() as model:

theta = pm.Beta("theta", alpha=5, beta=45)       # prior

obs  = pm.Binomial("obs", n=200, p=theta,       # likelihood

observed=30)

idata = pm.sample(2000, tune=1000, chains=4)     # posterior

import arviz as az

az.plot_posterior(idata)
```

A PyMC model describes **how the data could have been generated**: what I believe about the rate, and how observations arise given it. You write the story; the sampler infers. `tune=1000` discards the first 1,000 steps, while the sampler is still finding the high ground. `chains=4` runs four independent walks — which is what makes the next slide possible.

Did the Sampler Actually Work? (Martin et al. 2021)

MCMC can fail quietly. It always returns samples; whether they came from the posterior is a separate question.

Check	Want	What it means
$\hat{R}$	< 1.01	Did the chains land in the same place? If not, nothing else matters.
ESS	> 400	Effective sample size — 8,000 correlated draws may carry as much information as 200 independent ones.
Divergences	0	The sampler hit terrain it could not navigate. Even a few means the posterior is explored wrongly.

`az.summary(idata, ci_kind="hdi", ci_prob=0.94)` prints the interval, $\hat{R}$ and ESS. **Pass both arguments** — ArviZ now defaults to an 89% equal-tailed interval, not the HDI. On the model above: $\hat{R} = 1.00$, ESS 3,686, zero divergences. **Report all three.**

Prior Predictive Checks (McElreath 2020)

Before touching the data, simulate from the prior alone.

Two lines

```
with model:
prior_pred = pm.sample_prior_predictive(500)
az.plot_ppc(prior_pred, group="prior")
```

Ask whether the simulated data could plausibly have come from your business. Impossible values — negative sales, probabilities above one — mean the prior is misspecified, not merely wide. Data spanning a billion units means it is too vague; identical simulations mean it is too tight.

A prior predictive check is **model debugging** that costs two lines — and the habit most often skipped.

Key Takeaways

Bayesian inference: beliefs in, beliefs out. Data updates your prior into a posterior.

Lecture 10: Key Takeaways

1. **Bayesian inference** treats parameters as random variables with probability distributions. The posterior distribution $p(\theta \mid \mathcal{D})$ represents updated beliefs after seeing data.
2. **Bayes' theorem:** posterior $\propto$ likelihood $\times$ prior. The normalizing constant $p(\mathcal{D})$ is usually intractable, so we sample instead of computing it.
3. **Priors** encode prior knowledge. Use weakly informative priors (Beta, Normal, HalfNormal) to constrain parameters to plausible values without overwhelming the data.
4. **MCMC** (specifically NUTS in PyMC) draws samples from the posterior. From samples, compute any summary statistic: mean, median, HDI, or the probability that a parameter exceeds a threshold.
5. **Prior predictive checks** verify that your model generates plausible data before fitting. Run them every time.

Next: Bayesian Statistics II — Bayesian time series models, trend and seasonality, hierarchical models.

References I

Gelman, Andrew et al. (2013). *Bayesian Data Analysis*. 3rd. Boca Raton, FL: Chapman and Hall/CRC.

Martin, Osvaldo A., Ravin Kumar, and Junpeng Lao (2021). *Bayesian Modeling and Computation in Python*. Free online edition. Chapman and Hall/CRC. ISBN: 9780367894368. URL: https://bayesiancomputationbook.com/.

McElreath, Richard (2020). *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*. 2nd. Boca Raton, FL: Chapman and Hall/CRC.